Lecture Scribe Joint Random Variables

L15_L17 (1).pdf
PDF

Write a detailed lecture scribe for exam preparation based entirely on the given material. Carefully review the whole content and present every concept, definition, assumption, and step clearly. Explain how each idea is connected to the next and show the logical reasoning used in the lecture. Use only the information provided in the material. Do not add personal interpretations, outside knowledge, new examples, or extra explanations that are not included in the context. The final notes should be clear, organized, and easy to revise.

Here is a **detailed lecture scribe for exam preparation** created strictly from the provided lecture slides. All concepts, definitions, steps, and reasoning follow the exact structure of the lecture material.

Source:

Lecture Scribe

Joint Random Variables and Joint Distributions

(CSE400 – Fundamentals of Probability in Computing)

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1. Introduction: Why Study Joint Random Variables?

In science and real-life situations, we are often interested in **two or more random variables at the same time**. Instead of studying each variable separately, we study how they behave **together**. This requires understanding **joint random variables and their distributions**.

Examples

The lecture presents several situations where two variables are analyzed simultaneously:

- Height and weight of giraffes
- IQ and birthweight of children
- Frequency of exercise and rate of heart disease
- Air pollution level and respiratory illness rate
- Number of Facebook friends and age of members

These examples illustrate that multiple variables often interact with each other.

Additional Applications

Example 1: Smart Transportation

Two random variables are considered:

- **X = Vehicle Speed**
- **Y = Traffic Density**

Question:
Can vehicle speed be high when traffic density is also high?

This demonstrates that the variables can be **correlated**, meaning their values influence each other.

Applications in Intelligent Transportation Systems include:

- Adaptive traffic signals
- Congestion prediction
- Autonomous vehicle routing

Example 2: Wireless Network Performance

Two random variables:

- **X = Signal-to-Noise Ratio (SNR)**
- **Y = Data Throughput**

Throughput depends on SNR, but the relationship is affected by:

- Fading

- Interference
- Network congestion

Applications include:

- 5G scheduling
- Adaptive modulation
- Network planning

Example 3: Weather Prediction
Variables:

- **X = Rainfall**
- **Y = Temperature**

Used in weather forecasting.

Example 4: Drone Delivery
Variables:

- **X = Wind Speed**
- **Y = Delivery Time**

Used in logistics and UAV systems.

Example 5: Rolling Two Dice
Random variables:

- X = value on first die
- Y = value on second die

Questions like **X + Y = 7** or **X > Y** require understanding **joint probability**.

Purpose of Joint Distributions

Joint distributions allow us to:

- Compute probabilities involving **multiple variables**

- Understand the **relationship between variables**

When variables are **independent**, analysis is simpler.

When they are **not independent**, we measure dependence using:

- Covariance
- Correlation

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2. Discrete Case: Joint Probability Mass Function (Joint PMF)

Setup

Suppose two discrete random variables:

$X = \{x_1, x_2, \ldots, x_n\}$
$Y = \{y_1, y_2, \ldots, y_m\}$

The pair **(X, Y)** takes values in the **product set**:

$\{(x_1,y_1), (x_1,y_2), \ldots , (x_n,y_m)\}$

Definition: Joint PMF

The **joint probability mass function** is defined as:

$p(x_i, y_j) = P(X = x_i, Y = y_j)$

This represents the probability that:

- X takes value x_i
- Y takes value y_j

Both events occur simultaneously.

---

3. Joint PMF Table Representation

The joint PMF is commonly represented in a **table form**.

Rows represent values of **X**
Columns represent values of **Y**

Each cell contains:

$p(x_i, y_j)$

This table shows the probability for every possible combination of values.

Key Properties

1. Each probability must satisfy:

$0 \leq p(x_i, y_j) \leq 1$

2. The sum of all joint probabilities must equal 1:

$\Sigma \Sigma p(x_i, y_j) = 1$

---

4. Example: Rolling Two Dice

Experiment

Roll two fair dice.

Define:

X = value on the first die
T = total (sum) of both dice

Each outcome (i, j) occurs with probability:

1 / 36

Using this, we construct the **joint probability table of (X, T)**.

The table shows possible values of:

- X = 1 to 6
- T = 2 to 12

Each valid combination has probability **1/36**, while impossible combinations have probability **0**.

Observations

- Table entries are either **0 or 1/36**
- X and T are **not independent** random variables

This means the value of one variable affects the other.

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5. Continuous Case: Joint Probability Density Function (Joint PDF)

The continuous case is analogous to the discrete case.

The lecture highlights three key transitions:

1. **Discrete values → Continuous intervals**
2. **Joint PMF → Joint PDF**
3. **Summation → Double integrals**

Domain of the Variables

If:

$X \in [a, b]$
$Y \in [c, d]$

Then the pair (X, Y) lies in the **product region**:

$[a, b] \times [c, d]$

Definition: Joint PDF

A function **f(x, y)** is called the joint probability density function of X and Y if:

P((X,Y) lies in a small rectangle of area dx dy) ≈ f(x,y) dx dy

The diagram on page 18 illustrates a **small rectangle inside the region**, representing the probability area.

Key Properties

1. Non-negativity

f(x,y) ≥ 0

2. Total probability equals 1

∫∫ f(x,y) dx dy = 1 over the entire region

Important note:

The joint PDF **can be greater than 1**, because it represents **density**, not probability.

---

6. Continuous Joint PDF Example

Given

f(x,y) = 4xy

for

0 ≤ x ≤ 1
0 ≤ y ≤ 1

Thus, (X,Y) lies in the **unit square $[0,1]^2$**.

Tasks

1. Show that f(x,y) is a valid joint PDF

2. Visualize the event

$A = \{X < 0.5, Y > 0.5\}$

3. Compute P(A)

Step 1: Check Validity (Normalization)

Compute total probability:

$\int_0^1 \int_0^1 4xy \, dx \, dy$

First integrate with respect to x:

$\int_0^1 4xy \, dx = 2y$

Then integrate with respect to y:

$\int_0^1 2y \, dy = y^2 |_0^1 = 1$

Therefore:

- $f(x,y) \geq 0$
- Total probability = 1

Hence **f(x,y) is a valid joint PDF**.

Step 2: Compute P(A)

Event:

$A = \{X < 0.5, Y > 0.5\}$

Compute probability:

$P(A) = \int_0^{0.5} \int_{0.5}^1 4xy \, dy \, dx$

Integrating yields:

$P(A) = 3 / 16$

---

7. Joint Cumulative Distribution Function (Joint CDF)

Suppose X and Y are jointly distributed random variables.

We use the notation:

$X \leq x$, $Y \leq y$

to represent the event:

$\{X \leq x \text{ and } Y \leq y\}$.

Definition

The **joint cumulative distribution function** is defined as:

$F(x,y) = P(X \leq x , Y \leq y)$

This function gives the probability that both variables are less than or equal to specified values.

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8. Joint CDF for Continuous Case

If X and Y are continuous random variables with joint density f(x,y) over region $[a,b] \times [c,d]$, then:

$F(x,y) = \int\int f(u,v)\, du\, dv$

This integrates the joint density over the region:

$u \leq x$ and $v \leq y$.

Recovering the Joint PDF

The joint PDF can be obtained by differentiating the joint CDF:

$f(x,y) = \partial^2 F(x,y) / \partial x\, \partial y$

---

9. Joint CDF for Discrete Case

If X and Y are discrete random variables with joint PMF $p(x_i,y_j)$, then:

$F(x,y) = \Sigma \Sigma p(x_i,y_j)$

where:

$x_i \leq x$

$y_j \leq y$

Interpretation

To compute F(x,y):

1. Add probabilities from rows where $x_i \leq x$
2. Add probabilities from columns where $y_j \leq y$
3. Sum all corresponding entries in the joint probability table

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10. Properties of the Joint CDF

The joint CDF F(x,y) must satisfy the following properties.

1. Non-Decreasing Property

F(x,y) is **non-decreasing**.

If x or y increases, F(x,y) either:

- remains the same, or
- increases.

---

2. Lower-Left Corner Property

At the lower-left corner of the range:

$F(x,y) = 0$

If the corner is $(-\infty, -\infty)$:

$\lim F(x,y) = 0$

---

3. Upper-Right Corner Property

At the upper-right corner:

$F(x,y) = 1$

If the corner is (∞, ∞):

$\lim F(x,y) = 1$

Summary of the Lecture

This lecture introduces **joint random variables and joint distributions**.

Key topics include:

- Motivation for studying multiple random variables
- Joint PMF for discrete variables
- Joint PDF for continuous variables
- Example using dice
- Continuous PDF example with integration
- Joint cumulative distribution function
- Relationship between joint PDF and joint CDF
- Properties of the joint CDF

These concepts provide the foundation for analyzing **relationships between random variables** in probability and computing systems.

✅ If you want, I can also convert these **lecture scribes into a clean 10–15 page exam-ready PDF** (with headings, equations, and diagrams formatted like lecture notes).

Write a detailed lecture scribe for exam preparation based entirely on the given material. Carefully review the whole content and present every concept, definition, assumption, and step clearly. Explain how each idea is connected to the next and show the logical reasoning used in the lecture. Use only the information provided in the material. Do not add personal interpretations, outside knowledge, new examples, or extra explanations that are not included in the context. The final notes should be clear, organized, and easy to revise.

Here is a **complete lecture scribe for exam preparation** written strictly from the uploaded lecture slides. Every definition, assumption, concept, and step follows the lecture structure without adding outside information.

Source:

Lecture Scribe

Joint Random Variables and Joint Distributions

CSE400 – Fundamentals of Probability in Computing

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1. Motivation: Why Study Joint Random Variables?

In science and real-life situations, we are often interested in **two or more random variables at the same time**. Studying them together allows us to compute probabilities that involve both variables and to understand the **relationship between them**.

Examples where two variables are studied together include:

- Height and weight of giraffes
- IQ and birthweight of children

- Frequency of exercise and rate of heart disease
- Air pollution level and respiratory illness rate
- Number of Facebook friends and age of members

These examples show that many real-world situations involve **multiple variables that interact with each other**.

Example 1: Smart Transportation

Two random variables are considered:

- **X = Vehicle Speed**
- **Y = Traffic Density**

Question raised in the lecture:

Can speed be high when the traffic density is also high?

The discussion shows that the variables may be **correlated random variables**, meaning their values can depend on each other.

Applications mentioned:

- Adaptive traffic signals
- Congestion prediction
- Autonomous vehicle routing

Example 2: Wireless Network Performance

Two random variables:

- **X = Signal-to-Noise Ratio (SNR)**
- **Y = Data Throughput**

Throughput depends on SNR, but both variables vary due to:

- Fading
- Interference
- Congestion

Question raised:

If SNR is high, will throughput always be high?

The lecture explains that the relationship is influenced by network congestion, showing that variables may have **conditional relationships**.

Applications mentioned:

- 5G scheduling
- Adaptive modulation
- Network planning

Additional Examples

Further examples of joint variables presented in the lecture:

- Weather prediction
  - X = Rainfall
  - Y = Temperature
- Drone delivery
  - X = Wind Speed
  - Y = Delivery Time
- Rolling two dice
  - X and Y represent values on the dice

Questions like:

- $X + Y = 7$
- $X > Y$

require understanding the **joint probability mass function (joint PMF)**.

Importance of Joint Distributions

When random variables are studied together, they have a **joint distribution**.

This joint distribution allows us to:

- Compute probabilities of events involving both variables
- Understand the relationship between variables

When variables are **independent**, analysis becomes simpler.

When they are **not independent**, the dependence between them can be measured using:

- Covariance
- Correlation

---

2. Discrete Case – Joint PMF

Setup

Suppose **X and Y are discrete random variables**.

The values of the variables are defined as:

$X = \{x_1, x_2, \dots , x_n\}$

$Y = \{y_1, y_2, \dots , y_m\}$

The ordered pair **(X, Y)** takes values in the **product set**:

$\{(x_1,y_1), (x_1,y_2), \dots , (x_n,y_m)\}$

Thus every possible combination of X and Y values forms a possible outcome.

Definition: Joint PMF

The **joint probability mass function** is defined as:

$p(x_i, y_j) = P(X = x_i , Y = y_j)$

This represents the probability that:

- X takes the value x_i
- Y takes the value y_j

Both events occur simultaneously.

3. Joint PMF Table Representation

The joint PMF is often organized into a **table format**.

Structure of the table:

- Rows correspond to values of **X**
- Columns correspond to values of **Y**

Each cell in the table contains:

$p(x_i, y_j)$

This table shows the probability of each joint outcome.

Key Properties of the Joint PMF

1. Probability bounds:

$0 \leq p(x_i, y_j) \leq 1$

2. Total probability condition:

The sum of all joint probabilities must equal 1.

$$\sum_{i=1}^{n} \sum_{j=1}^{m} p(x_i, y_j) = 1$$

---

4. Example: Rolling Two Dice

Experiment

Roll two fair dice.

Define the random variables:

- **X = value on the first die**

- **T = total (sum) of both dice**

Each pair outcome (i, j) occurs with probability:

1 / 36

Using these probabilities, we construct the **joint probability table of (X, T)**.

The table includes:

- X values from 1 to 6
- T values from 2 to 12

Some combinations are possible and receive probability **1/36**, while impossible combinations receive probability **0**.

Observations from the Table

Two observations are stated in the lecture:

1. Entries are either **0 or 1/36**
2. X and T are **not independent**

This means the value of T depends on the value of X.

---

5. Continuous Case – Joint PDF

The continuous case is described as **analogous to the discrete case**.

The lecture shows the following correspondences:

- Discrete values → Continuous intervals
- Joint PMF → Joint PDF
- Summations → Double integrals

Domain of the Variables

If:

X ∈ [a, b]
Y ∈ [c, d]

Then the pair (X, Y) takes values in the **product region**:

[a, b] × [c, d]

Definition: Joint PDF

A function **f(x, y)** is called the **joint probability density function** of X and Y if:

P((X,Y) lies in a small rectangle of area dx dy) ≈ f(x,y) dx dy

The lecture illustrates this using a rectangle inside the region where:

Probability = f(x,y) dx dy.

Key Properties

1. Non-negativity:

f(x,y) ≥ 0

2. Total probability equals 1:

$$\int_c^d \int_a^b f(x, y)\, dx\, dy = 1$$

An important note stated in the lecture:

The joint PDF **may take values greater than 1** because it represents a **density**, not a probability.

---

6. Continuous Joint PDF Example

Given

f(x,y) = 4xy

with:

0 ≤ x ≤ 1
0 ≤ y ≤ 1

Therefore, the pair (X, Y) lies in the **unit square**:

$[0,1]^2$

Tasks

1. Show that f(x,y) is a valid joint PDF
2. Visualize the event

A = {X < 0.5 , Y > 0.5}

3. Compute P(A)

Step 1: Check Validity (Normalization)

Compute:

$$\int_0^1 \int_0^1 4xy \, dx \, dy$$

First integrate with respect to x:

$$\int_0^1 4xy \, dx = 2y$$

Then integrate with respect to y:

$$\int_0^1 2y \, dy = y^2 \Big|_0^1 = 1$$

Thus:

- f(x,y) ≥ 0
- Total probability = 1

Therefore **f(x,y) is a valid joint PDF**.

Step 2: Compute P(A)

Event:

A = {X < 0.5 , Y > 0.5}

Compute probability:

$$P(A) = \int_0^{0.5} \int_{0.5}^{1} 4xy \, dy \, dx$$

After evaluating the integrals:

$$P(A) = \frac{3}{16}$$

---

7. Joint Cumulative Distribution Function (Joint CDF)

Suppose X and Y are jointly distributed random variables.

Notation used:

X ≤ x , Y ≤ y

This represents the event:

{X ≤ x and Y ≤ y}.

Definition

The **joint cumulative distribution function** is defined as:

F(x,y) = P(X ≤ x , Y ≤ y)

This gives the probability that both random variables are less than or equal to specified values.

---

8. Joint CDF – Continuous Case

If X and Y are continuous random variables with joint density f(x,y) over region [a,b] × [c,d], then the joint CDF is obtained by integrating the density:

$$F(x,y) = \int_c^y \int_a^x f(u,v)\,du\,dv$$

Recovering the Joint PDF

Since there are two variables, we use partial derivatives.

The joint PDF can be obtained from the joint CDF by:

$$f(x,y) = \frac{\partial^2}{\partial x \partial y} F(x,y)$$

---

9. Joint CDF – Discrete Case

If X and Y are discrete random variables with joint PMF $p(x_i, y_j)$, the joint CDF is obtained by summing probabilities:

$$F(x,y) = \sum_{x_i \le x} \sum_{y_j \le y} p(x_i, y_j)$$

Interpretation

To compute F(x,y):

1. Add all probabilities in the table
2. Include rows where $x_i \leq x$
3. Include columns where $y_j \leq y$

---

10. Properties of the Joint CDF

The joint CDF F(x,y) must satisfy several properties.

Property 1: Non-Decreasing

F(x,y) is **non-decreasing**.

If x or y increases, then F(x,y) must either:

- stay constant, or
- increase.

Property 2: Lower-Left Corner

At the lower-left corner of the joint range:

$F(x,y) = 0$

If the lower-left corner is $(-\infty, -\infty)$, then:

$\lim F(x,y) = 0$.

Property 3: Upper-Right Corner

At the upper-right corner of the joint range:

$F(x,y) = 1$

If the upper-right corner is (∞, ∞), then:

lim F(x,y) = 1.

Final Summary of the Lecture

This lecture develops the concept of **joint random variables** and their distributions.

The main ideas are:

1. Motivation for studying multiple random variables together
2. Joint PMF for discrete variables
3. Joint PMF table representation and its properties
4. Example using two dice
5. Joint PDF for continuous variables
6. Continuous joint density example with normalization and probability calculation
7. Definition of the joint cumulative distribution function
8. Joint CDF for continuous variables using integration
9. Joint CDF for discrete variables using summation
10. Important properties of the joint CDF

These concepts form the basis for analyzing **relationships between multiple random variables in probability theory and computing applications**.

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Lecture Scribe

Joint Random Variables and Joint Distributions

Course: CSE400 – Fundamentals of Probability in Computing
Lectures: L15–L18

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1. Introduction: Why Study Joint Random Variables?

In science and real life, we are often interested in **two or more random variables at the same time**. Studying these variables together allows us to compute probabilities involving both variables and understand the relationship between them.

Examples of situations involving two variables

The lecture lists several examples where two variables are considered simultaneously:

- Height and weight of giraffes
- IQ and birthweight of children
- Frequency of exercise and rate of heart disease
- Air pollution level and respiratory illness rate
- Number of Facebook friends and age of members

These examples demonstrate that many real-world problems involve **multiple variables whose behaviors are related**.

Example 1: Smart Transportation

In a smart traffic system two random variables are defined:

X = Vehicle Speed
Y = Traffic Density

The lecture raises the question:

Can speed be high when traffic density is also high?

This question highlights that the variables may be **correlated random variables**, meaning their values may influence each other.

Applications mentioned include:

- Adaptive traffic signals
- Congestion prediction
- Autonomous vehicle routing

Example 2: Wireless Network Performance

Two random variables are defined:

X = Signal-to-Noise Ratio (SNR)
Y = Data Throughput

Throughput depends on SNR, but both variables vary due to:

- Fading
- Interference
- Congestion

The lecture asks:

If SNR is high, will throughput always be high?

The reasoning provided is that throughput may not always be high due to network congestion, showing that the variables have **conditional relationships**.

Applications mentioned:

- 5G scheduling
- Adaptive modulation
- Network planning

Additional Examples

The lecture also presents other examples of joint variables:

Weather Prediction
X = Rainfall
Y = Temperature

Drone Delivery
X = Wind Speed
Y = Delivery Time

Rolling Two Dice
X and Y represent values obtained when two dice are rolled.

Questions such as:

$X + Y = 7$
$X > Y$

require the use of **joint probability mass functions**.

Importance of Joint Distributions

When random variables are considered together, they have a **joint distribution**.

The joint distribution allows us to:

- Compute probabilities of events involving both variables
- Understand the relationship between the variables

When variables are **independent**, the analysis becomes simpler.

When they are **not independent**, the nature of dependence can be measured using:

- Covariance
- Correlation

---

2. Discrete Case – Joint Probability Mass Function (Joint PMF)

Setup

Suppose X and Y are **discrete random variables**.

The possible values are:

$X = \{x_1, x_2, \dots, x_n\}$
$Y = \{y_1, y_2, \dots, y_m\}$

The ordered pair (X, Y) takes values in the **product set**:

$\{(x_1,y_1), (x_1,y_2), \dots, (x_n,y_m)\}$

Thus every possible combination of values of X and Y represents a possible outcome.

Definition: Joint PMF

The **joint probability mass function** is defined as:

$p(x_i, y_j) = P(X = x_i, Y = y_j)$

This gives the probability that:

X takes value x_i
and
Y takes value y_j

simultaneously.

---

3. Joint PMF Table Representation

The joint PMF can be organized in a **table format**.

Structure of the table:

- Rows represent values of X
- Columns represent values of Y

Each cell in the table contains:

$p(x_i, y_j)$

This representation shows the probability of each joint outcome.

Key Properties of the Joint PMF

1. Probability bounds:

$0 \leq p(x_i, y_j) \leq 1$

2. Total probability condition:

$$\sum_{i=1}^{n} \sum_{j=1}^{m} p(x_i, y_j) = 1$$

This means the sum of probabilities of all possible outcomes must equal 1.

---

4. Example: Rolling Two Dice

Experiment

Roll two fair dice.

Define random variables:

X = value on the first die
T = total (sum) of both dice

Each possible pair outcome (i, j) has probability:

1/36

Using this information, the **joint probability table of (X, T)** is constructed.

Possible values:

X = 1, 2, 3, 4, 5, 6
T = 2, 3, 4, … , 12

Some combinations are possible and have probability **1/36**, while impossible combinations have probability **0**.

Observations from the Table

The lecture states two observations:

1. Entries in the table are either **0 or 1/36**
2. X and T are **not independent**

This indicates that the value of T depends on the value of X.

---

5. Continuous Case – Joint Probability Density Function (Joint PDF)

The continuous case is described as **analogous to the discrete case**.

The lecture presents the following correspondences:

Discrete values → Continuous intervals
Joint PMF → Joint PDF
Summations → Double integrals

Domain of the Variables

If:

X ∈ [a, b]
Y ∈ [c, d]

then the pair (X, Y) takes values in the **product region**:

[a,b] × [c,d]

Definition: Joint PDF

A function **f(x,y)** is called the **joint probability density function** of X and Y if:

P((X,Y) lies in a small rectangle of area dx dy) ≈ f(x,y) dx dy

This means the probability that (X,Y) lies within a small rectangle around (x,y) is approximately equal to the density multiplied by the rectangle area.

Key Properties

1. Non-negativity:

f(x,y) ≥ 0

2. Total probability equals 1:

$$\int_c^d \int_a^b f(x,y)\,dx\,dy = 1$$

The lecture also states that:

The joint PDF may take values **greater than 1**, because it represents **density rather than probability**.

---

6. Continuous Joint PDF Example

Given

f(x,y) = 4xy

with

0 ≤ x ≤ 1
0 ≤ y ≤ 1

Thus the pair (X,Y) lies in the **unit square [0,1]²**.

Tasks in the Example

1. Show that f(x,y) is a valid joint PDF
2. Visualize the event

A = {X < 0.5 , Y > 0.5}

3. Compute P(A)

Step 1: Check Validity (Normalization)

Compute:

$$\int_0^1 \int_0^1 4xy \, dx \, dy$$

First integrate with respect to x:

$$\int_0^1 4xy \, dx = 2y$$

Then integrate with respect to y:

$$\int_0^1 2y \, dy = y^2 \Big|_0^1 = 1$$

Thus:

- f(x,y) ≥ 0
- Total probability = 1

Therefore **f(x,y) is a valid joint PDF**.

Step 2: Compute P(A)

Event:

A = {X < 0.5 , Y > 0.5}

Probability is computed as:

$$P(A) = \int_0^{0.5} \int_{0.5}^{1} 4xy \, dy \, dx$$

After evaluating the integrals:

$$P(A) = \frac{3}{16}$$

---

7. Joint Cumulative Distribution Function (Joint CDF)

Suppose X and Y are **jointly distributed random variables**.

The notation:

X ≤ x , Y ≤ y

represents the event:

{X ≤ x and Y ≤ y}.

Definition

The **joint cumulative distribution function** is defined as:

F(x,y) = P(X ≤ x , Y ≤ y)

This gives the probability that both random variables are less than or equal to specified values.

---

8. Joint CDF – Continuous Case

If X and Y are continuous random variables with joint density f(x,y) over region [a,b] × [c,d], then the joint CDF is obtained by integrating the density:

$$F(x, y) = \int_c^y \int_a^x f(u, v)\, du\, dv$$

Recovering the Joint PDF

Since there are two variables, **partial derivatives** are used.

The joint PDF can be obtained from the joint CDF by:

$$f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$$

---

9. Joint CDF – Discrete Case

If X and Y are discrete random variables with joint PMF $p(x_i, y_j)$, the joint CDF is obtained by summing probabilities:

$$F(x, y) = \sum_{x_i \le x} \sum_{y_j \le y} p(x_i, y_j)$$

Interpretation

To compute F(x,y):

- Add all probabilities in the table
- Include rows with $x_i \leq x$
- Include columns with $y_j \leq y$

---

10. Properties of the Joint CDF

The joint CDF F(x,y) must satisfy several properties.

Property 1: Non-Decreasing

F(x,y) is **non-decreasing**.

If x or y increases, F(x,y) must either:

- remain constant, or
- increase.

Property 2: Lower-Left Corner

At the lower-left corner of the joint range:

F(x,y) = 0

If the lower-left corner is (−∞, −∞):

$$\lim_{(x,y)\to(-\infty,-\infty)} F(x,y) = 0$$

Property 3: Upper-Right Corner

At the upper-right corner of the joint range:

F(x,y) = 1

If the upper-right corner is (∞, ∞):

$$\lim_{(x,y)\to(\infty,\infty)} F(x,y) = 1$$

Final Summary of the Lecture

This lecture develops the concept of **joint random variables and joint distributions**.

The main topics covered are:

- Motivation for studying multiple random variables together
- Joint probability mass function for discrete variables
- Joint PMF table representation and its properties
- Example using two dice
- Joint probability density function for continuous variables
- Continuous joint PDF example with normalization and probability calculation
- Definition of the joint cumulative distribution function
- Joint CDF for continuous and discrete cases
- Important properties of the joint CDF

These concepts provide the foundation for analyzing **probabilities involving more than one random variable**.

Is this conversation helpful so far?